

```
#include <iostream>
```

```
int main() {
```

```
    double appliance_power, total_load = 0.0;
```

```
    const double MAX_LIMIT = 5000.0;
```

```
    std::cout << "Load Testing Aggregation System Module (Up to 20 Appliance Inputs):\n";
```

```
    for (int i = 1; i <= 20; ++i) {
```

```
        std::cout << "Enter power rating value for appliance component item #" << i << "\n (W): ";
```

```
        std::cin >> appliance_power;
```

```
        total_load += appliance_power;
```

```
        if (total_load > MAX_LIMIT) {
```

```
            std::cout << "Warning Threshold Triggered: Maximum load exceeded. Stop adding appliances." << std::endl;
```

```
            break;
```

```
        }
```

```
    }
```

```
    std::cout << "\n--- Load Test System Reporting Terminal ---\n";
```

```
    std::cout << "Final structural design aggregate total load metric calculation: " << total_load << " W" << std::endl;
```

```
    return 0;
```

```
}
```